Solutions

A12-M01

0 and 2; the matches overlap.

A12-M02

0, 1, 2, 3, 4, the five boundaries.

A12-M03

The empty set, because the pattern is longer than the text.

A12-M04

$\pi = [0,0,1,2,3,0,1]$. At $i=5$, $k=3 \to 1 \to 0$ while comparing c with b, b, a; all fail.

A12-M05

For states 0,1,2,3, transitions (a,b) are (1,0), (1,2), (3,0), (1,2).

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A12-M06

Yes. $12 \bmod 13 = 25 \bmod 13 = 12$, but the strings differ, so verify hash hits character by character.

A12-M07

12, 10, 8, the residues of 12, 23, 34 modulo 13.

A12-M08

0, 1, 2. $\pi = [0,1,2]$, and after each match fall back to 2.

A12-M09

Each symbol increases state by at most 1; each fallback decreases it by at least 1. State is nonnegative, so total fallbacks cannot exceed increases.

A12-M10

Verify whenever a window hash equals the pattern hash; normalize the remainder to 0..q−1. Let m=⌊n/2⌋. All n−m+1 windows truly match, so each requires m comparisons: m(n−m+1)=Θ(nm), despite zero false hits. For $1 \le m \le n$, if H counts all true and false hash hits, the bound is O(n+m+mH), with worst case Θ(n+m+m(n−m+1)). With V true matches and a randomized hash with collision probability at most 1/q for each unequal window, expected work is O(n+m+mV+mn/q).

A12-X01

π=[0,0,1]. At q=3=m, do not access P[3]; fall back to π[2]=1. Since δ(1,a)=1 and δ(1,b)=2, δ(3,a)=1 and δ(3,b)=2. Directly, abaa has longest pattern-prefix suffix a, and abab has ab. Both suffixes can still be extended, preserving a possible overlapping match.

A12-X02

π=[0,0,1,2,3,0,1]. Start with the base case π[0]=0 and build the remaining entries from i=1. Starting at 5, π[4]=3, π[2]=1, π[0]=0, so the proper border lengths are 3,1,0. Every shorter border is a border of the longest one, and conversely, so recursion lists all of them. x differs from P[5]=c, then 5→3 and x differs from P[3]=b, then 3→1 and x differs from P[1]=b, then 1→0 and x differs from P[0]=a; the state remains 0.

A12-X03

At b=10 the hashes are 1,0,1,9,1. The 12 windows at positions 0 and 4 are true matches; 23 at position 2 is a false hit and is rejected by verification, so the output is {0,4}. The nonmatching windows 22,23,31 collide with P at b=0,10,6, respectively, each with probability 1/11. Linearity gives E[f]=3/11.

A12-X04

The nonzero difference polynomial has degree at most 4, hence at most 4 roots in F_{11}. Each nonmatching window collides with probability at most 4/11. If f is the sum of three collision indicators, $E[f] \le 12/11$ and expected character comparisons for verification are at most 5z+5 · 12/11. Independence is unnecessary for linearity of expectation; this is an upper bound, and a comparison can stop at the first differing character.

References and attribution

CLRS, Introduction to Algorithms, 3rd ed., chapter 32, pp. 990–1006. MIT 6.046J Fall 2004, Lecture 22.

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